Here, the total number of square chocolate pieces is 20, which represents the area of the whole chocolate bar.

The total number of squares in all these cases can be arrived by multiplying the number of squares along its length with the number of squares along its breadth instead of counting them.

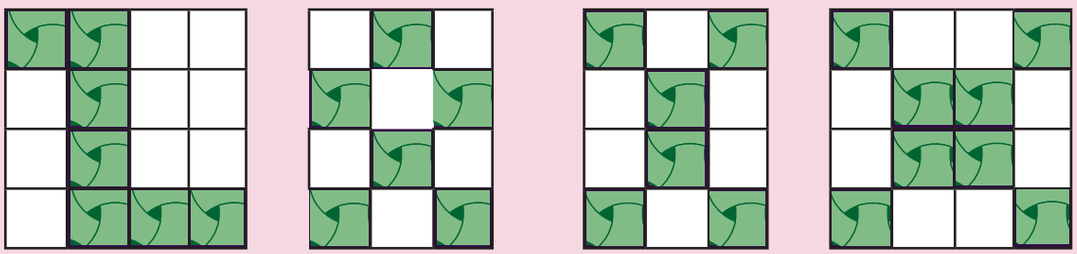
Note

'Square units' can also be written as 'unit^2'.

Therefore the area of any rectangle = (length x breadth) square units.
= (l x b) sq. units.

Try these

Find the number of tiles required to fill the area of following figures.



i) ii) iii) iv)

Example 9 Find the area of a rectangle of length 12 *cm* and breadth 7 *cm*.

Solution

Length of the rectangle, l = 12 *cm.*
Breadth of the rectangle, b = 7 *cm.*
Area of the rectangle A = (l x b) *sq. units.*
= 12 x 7 = 84 *sq. cm.*

3.3.2 Area of a Square

If the length and breadth of a rectangle are equal, then it becomes a square.

Area of the rectangle = (length x breadth) square units.
= (side x side) *sq. units.*
= (s x s) *sq. units.*
= Area of a square

Therefore area of a square = (s x s) *sq. units.*

Note

When the rectangle is made into a square then length (l)=breadth (b)=side (s)

Example 10 Find the area of a square of side 15 *cm.*

Solution

Side of the square, s = 15 *cm*
Area of the square, A = (s x s) *sq. units.*
= 15 x 15
= 225 sq. *cm*. (or) 225 cm^2

3.3.3 Area of a Right Angled Triangle

In a right angled triangle one of the sides containing the right angle is treated as its base (b units) and the other side as its height (h units).

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